



ENGINEERING PORTFOLIO

ESHWAR PAMULA
THE OHIO STATE UNIVERSITY



Flux Marine – Powertrain Engineering

Note: Due to NDA restrictions, I cannot disclose all technical projects or proprietary information from my work at Flux.

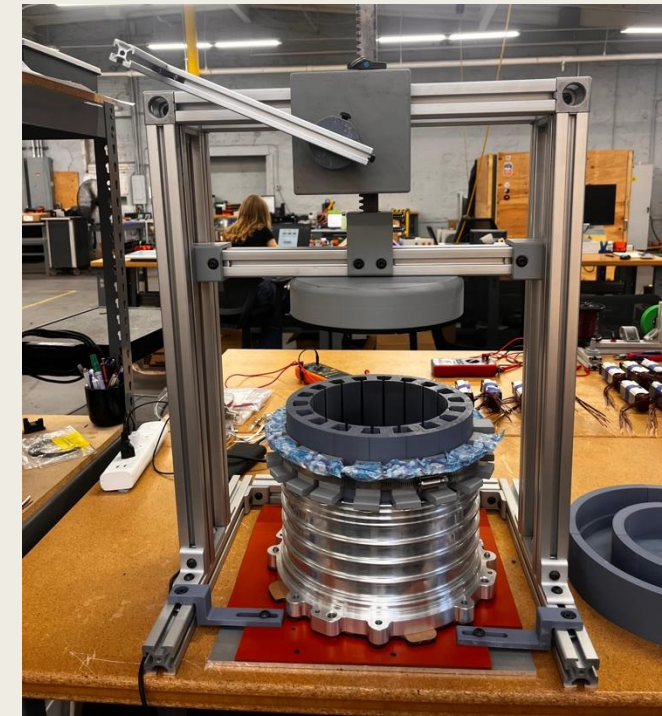
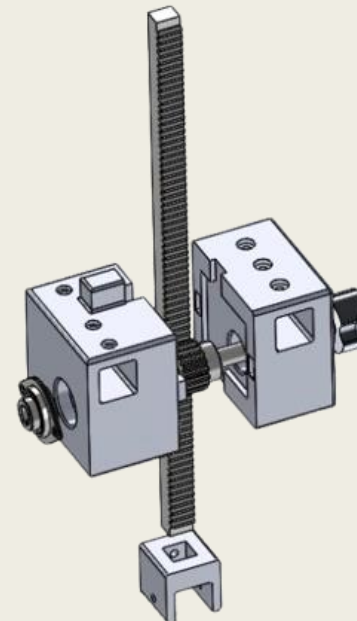
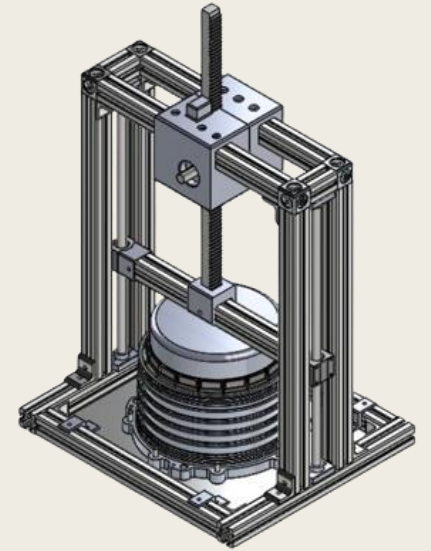
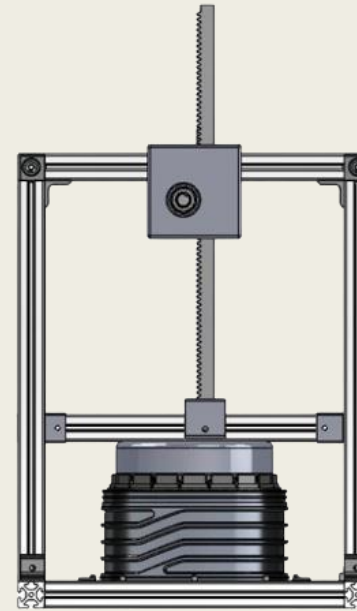
Helped build the stator (segmented windings, assembly, and potting) and rotor (magnet assembly and carbon fiber wrap) for the R&D motor built in house.

Shrink-Fit Assembly Jig: I designed and built a custom arbor press system for shrink-fitting operations.

Ran Dynamometer tests for the electric motor.

Designed and built parts and to be used on the Flux's outboard assembly line for Highfield and Scout boats.

Helped design and build fixtures for testing the updated battery packs for the Gen 2 outboard.



Allegion – R&D Engineering

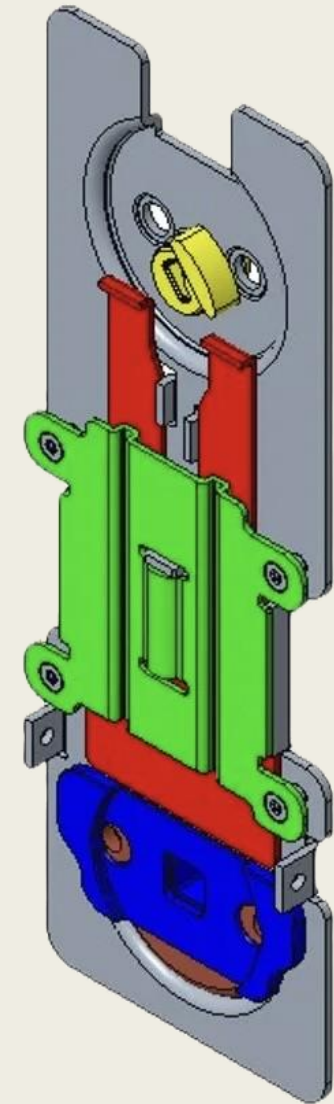
As a part of my summer internship as a mechanical engineering intern at Allegion, I was tasked to develop a series of lock integration solutions (mechanical and electronic) and develop a new Apollo Interconnect Slider Asset.

Worked on Two New Products:

1. Schlage B560F with Indicator and IC Slider Asset
2. Schlage Encode and IC Slider Asset

Design an interconnect system that would solve a frustration in residential security: the disconnect between traditional mechanical door locks and modern electronic touchpad locks.

Traditionally, homeowners have had to operate two separate locking mechanisms: the standard mechanical deadbolt that comes with their home, and an additional electronic touchpad lock they might install later. This creates a cumbersome experience, requiring users to operate two different locks.



Allegion – R&D Engineering

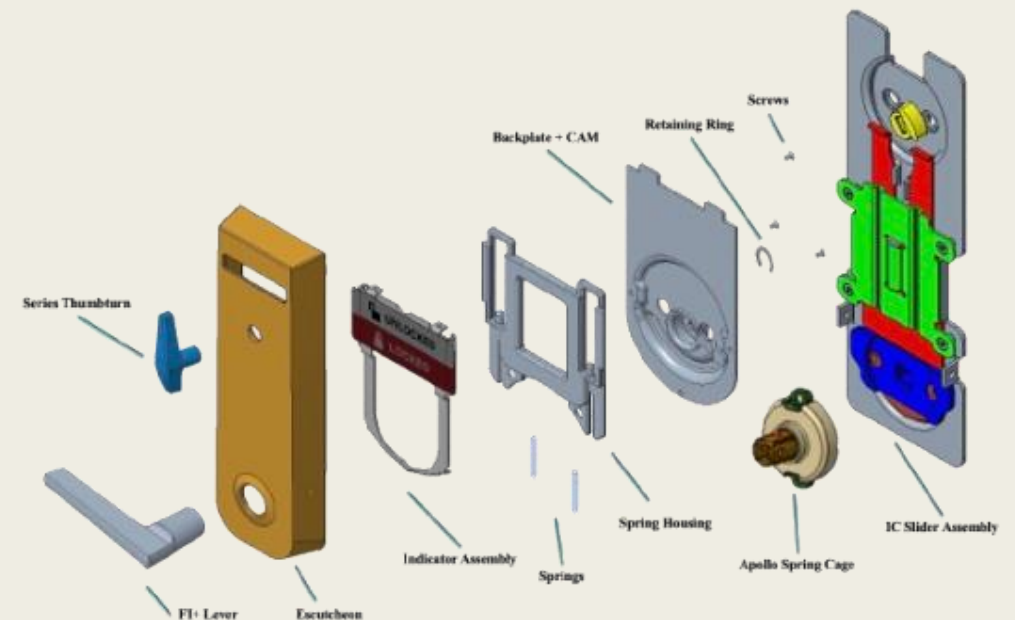
Proposed Solution:

- From outside the home: When the electronic lock is unlocked via code, card, or smartphone, the mechanism simultaneously unlocks the traditional deadbolt.
- From inside the home: A single action of pulling down the interior lever unlocks both the electronic and mechanical locks together.

The double-CAM design was critical to enabling smooth, reliable mechanical transfer between the two lock types, creating a single security solution.

The left two images shows locked vs unlocked position.

Note: The thumb turn is press-fit with during assembly and does not hold its position here. The latch does not retract all the way back due to using 3D printed CAMs.



Formula Buckeyes

FSAE – Suspension

Have been on the suspension team of Formula Buckeyes FSAE for 2.5 years.

Helped secure 3rd place overall at FSAE Michigan 2023, our highest finish since 2001, followed by a 1st place overall in 2024.

Machined uprights, mounts, and suspension hard points using mill and lathe operations. Designed and implemented adjustable anti-roll bar systems for both front and rear suspensions.

- Front Upright: Machined from 7075-T6 aluminum, weighing only 0.93 lbs.
- Rear Upright: Matched design language with specific adaptations for rear suspension geometry, weighing 1.00 lbs.



Formula Buckeyes FSAE – Suspension

Anti-Roll Bar Design and Manufacture

I designed adjustable anti-roll bar (ARB) systems for both front and rear suspensions to fine-tune the vehicle's handling characteristics and lateral load transfer distribution.

1. Front and rear ARBs feature multiple adjustment points to modify lateral load transfer distribution
2. Chassis design was analyzed to ensure torsional stiffness did not compromise ARB effectiveness
3. Three front inboard mounting positions for additional tuning capability for load transfer and anti-dive characteristics

Here is the [link](#) for a detailed analysis of the design of the ARB.

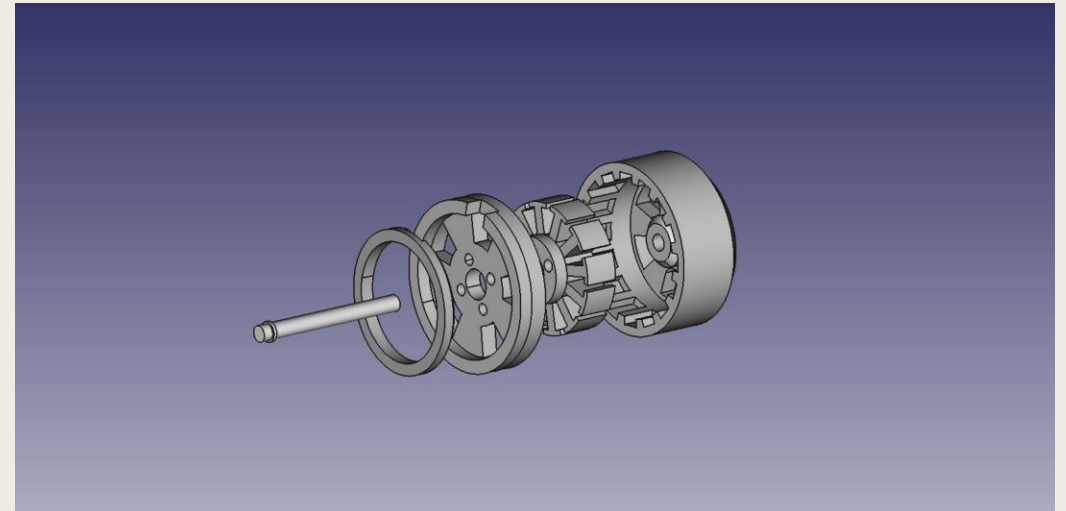
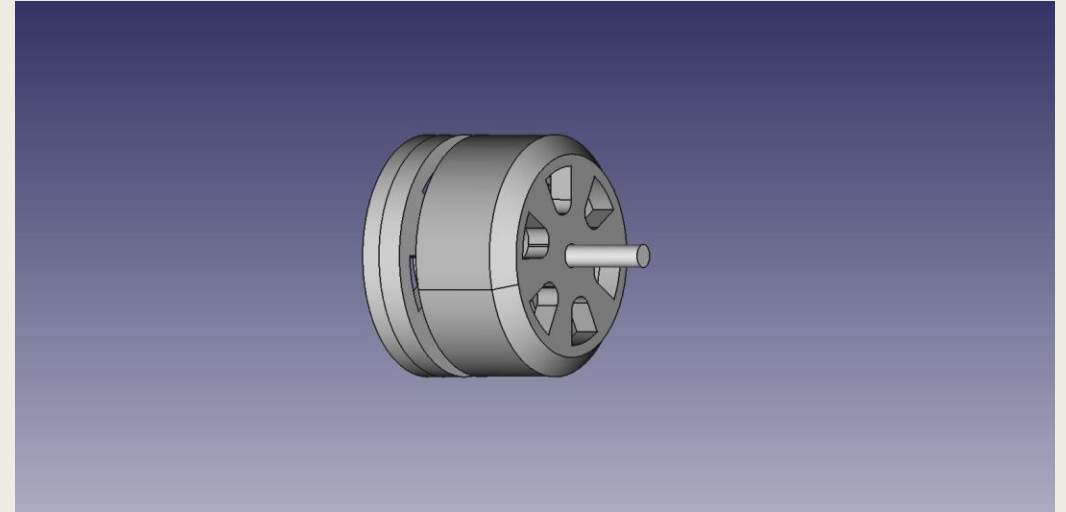


Brushless Mini Motor

I wanted to attempt to build a brushless DC mini motor from scratch.

Bill of Materials:

1. 5mm diameter metal rod.
2. M5 Die (for threading)
3. 4 x MR95ZZ bearings (9 x 5 x 3 mm)
4. 4 x M3 brass inserts
5. 4 x M3 screws
6. 0.5mm copper wire (enough length for 18 x 20 turns)
7. Magnets (10 x 5 x 2 mm)



Brushless Mini Motor

Building the Rotor:

Alternate between North and South Poles on each of the magnet within the rotor.

I used JB Weld epoxy to keep the magnets firm in place.

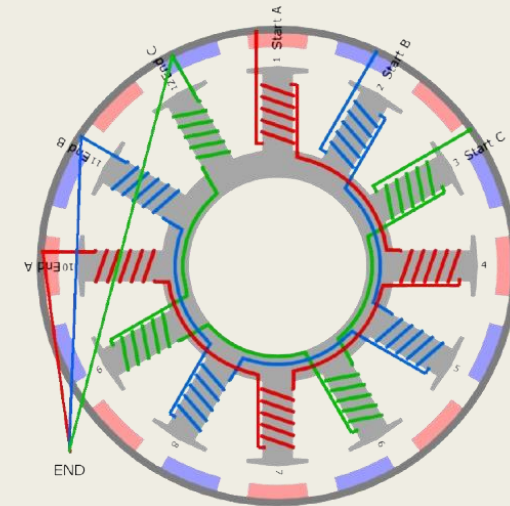
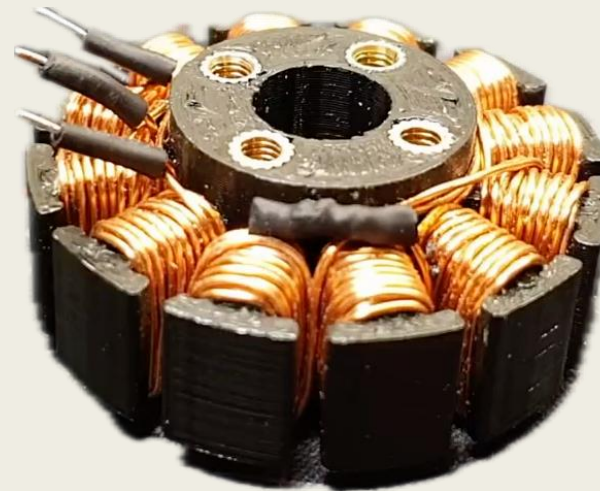
Ended up using too much epoxy for the first build and failed to remove excess and had to rebuild the rotor.

Building the Stator:

All coils must be wound in the same direction at every third slot.

I used a Y configuration for winding.

The three ends should be connected to each other and beginning to the ESC.



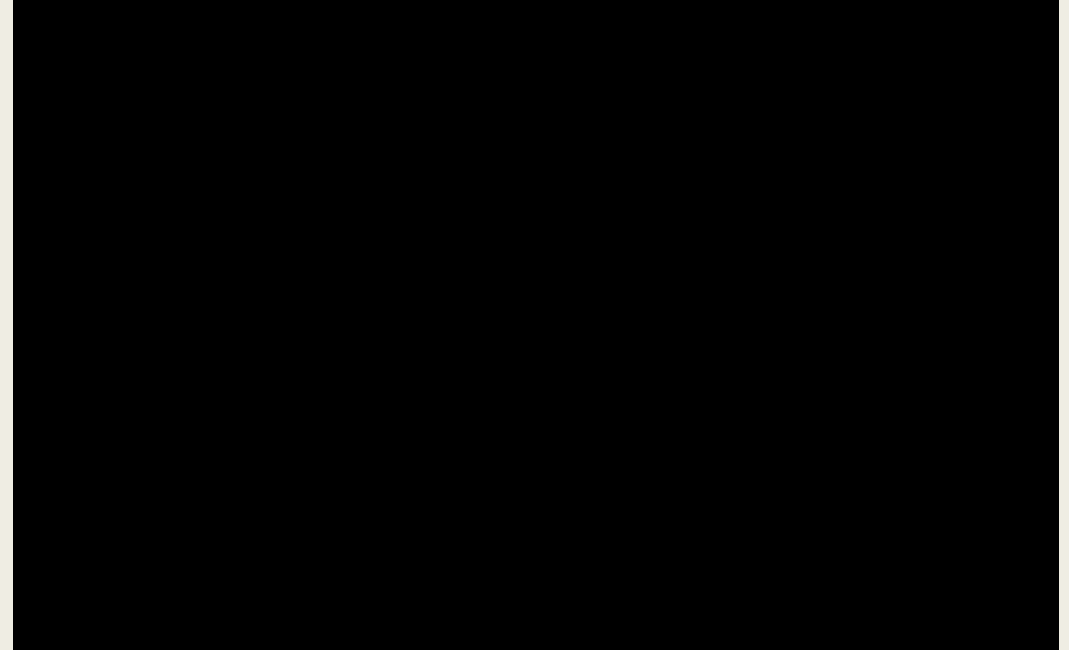
Brushless Mini Motor

Result:

The motor did run but needed a slight push at the start.

I believe to be because of the 3D print and/or the magnets having too much of a separation within the rotor.

For my next build, I hope to build a full-size axial flux motor.



Arduino Projects

Smart home automation built with Arduino

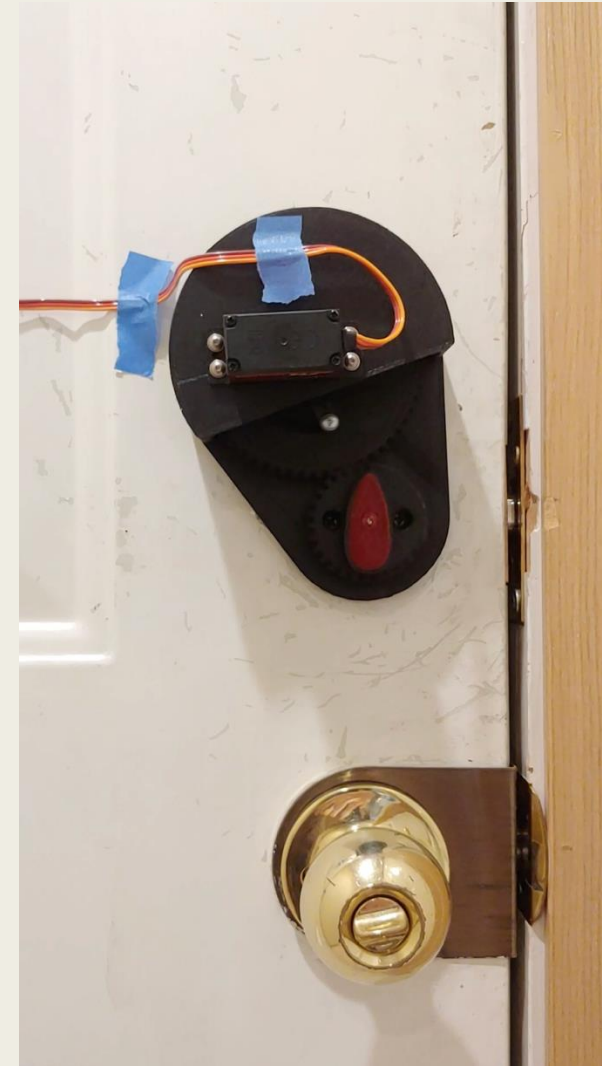
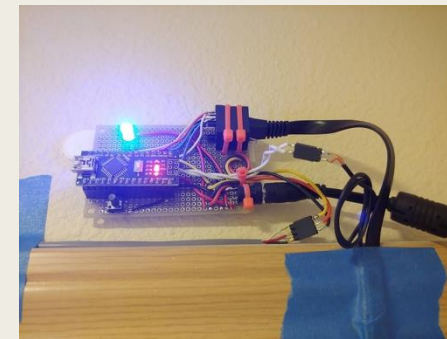
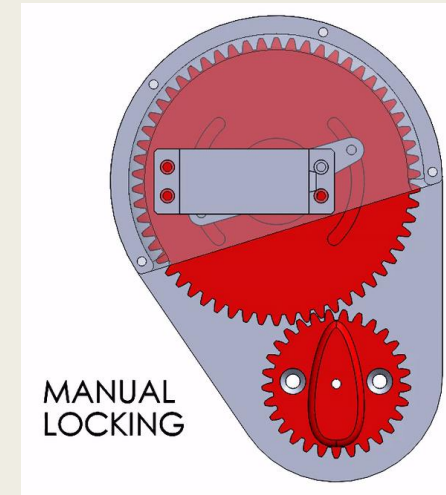
RFID Door Locking System:

Developed an automatic RFID door locking system for my room during Summer 2024, integrating both hardware and software components into one solution.

The design replaces only the interior half of a standard Schlage deadbolt assembly, making use of the original mounting screws for a clean installation.

The system features RFID access, allowing new authentication cards to be easily added or removed from the Arduino's EEPROM using a designated master card. Key components include:

- Arduino Nano microcontroller
- MFRC522 RFID reader module
- Continuous rotation servo motor
- Custom 3D-printed mounting bracket
- Power management circuit with sleep mode for extended battery life



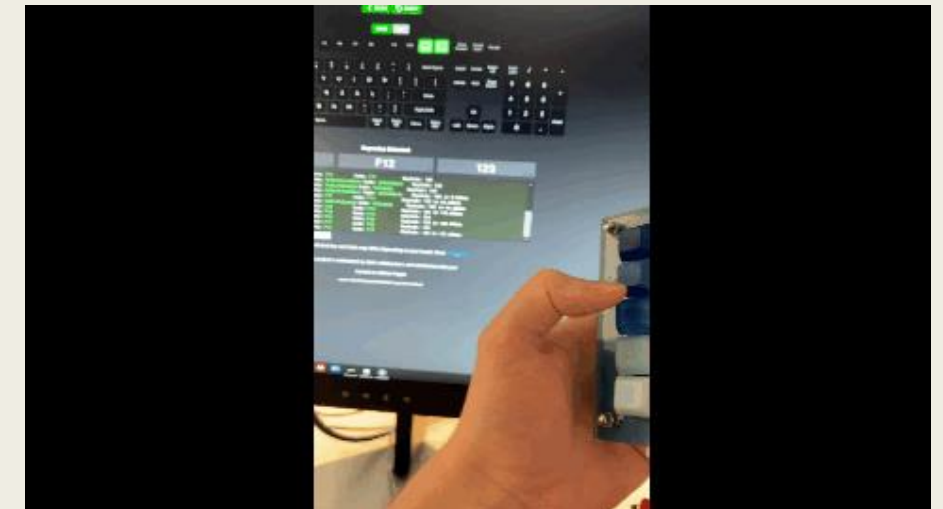
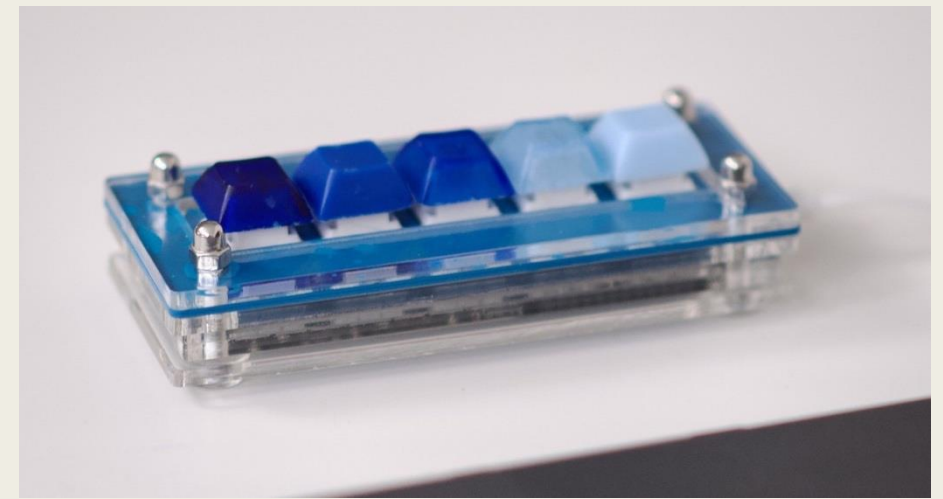
5-Key Mini Keyboard

I took up this personal project in my garage from ideation to production after coming across a similar concept on Amazon. The idea was to make a mini keyboard with 5 of my most frequently used keys: Space, Enter, Escape, Shift, and Delete.

The keyboard uses an Arduino Pro Micro (ATmega32U4) microcontroller which natively supports USB HID functionality.

The keycaps are custom PBT double-shot with side-printed legends for durability and a premium feel. The case consists of a 5-layer stacked acrylic design (3mm thickness per layer) which provides structural integrity while showcasing the internal components.

To connect everything, I designed a custom single-layer PCB with through-hole components and added a USB-C port with ESD protection circuit for modern connectivity. The firmware is a modified version of QMK, allowing for custom key mapping and macros.



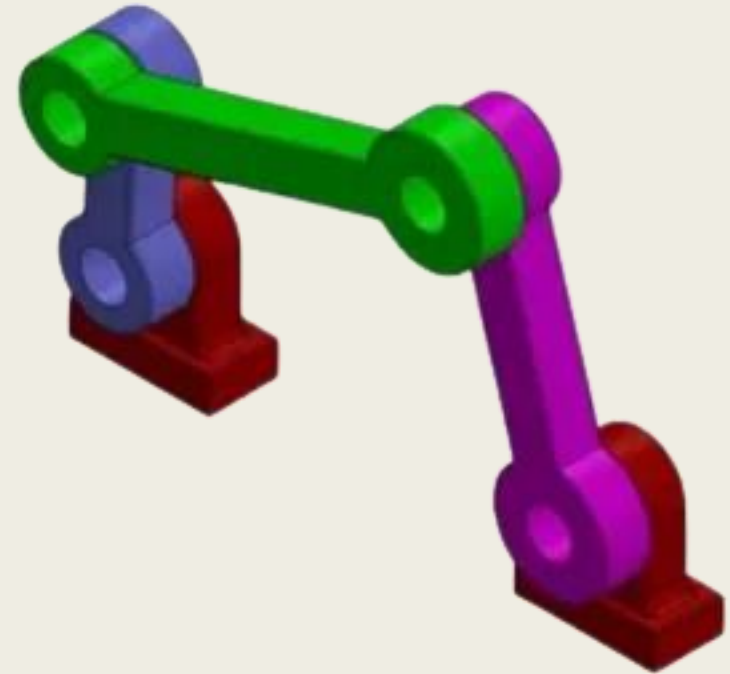
Four Bar Motion Study

The goal of this project was to assemble a Four-Bar Mechanism with provided parts and perform a motion study to observe how the parts interact with each other, obtaining quantitative data on metrics such as torque, power, and reaction forces.

Mechanism Components:

The assembly consists of two bearings and three bars. The two fixed bearings can be replaced with one fixed bar without affecting the function of the mechanism, thus the name "four bar". The main parts are:

- Bearing 1: Red part on the left (fixed)
- Bearing 2: Red part on the right (fixed)
- Crank: Blue part Extension:
- Green part Rocker: Magenta Part



Four Bar Motion Study

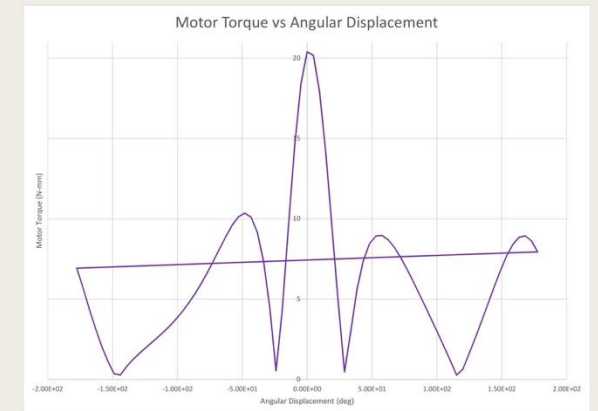
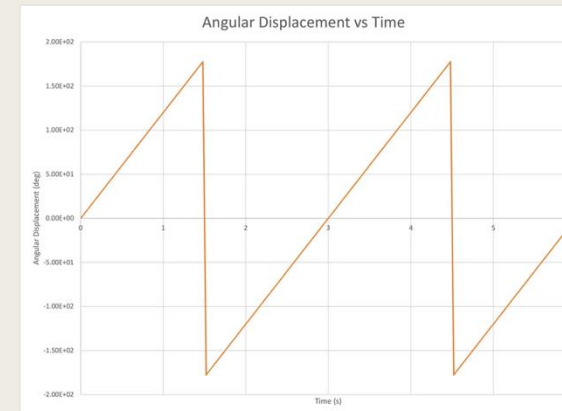
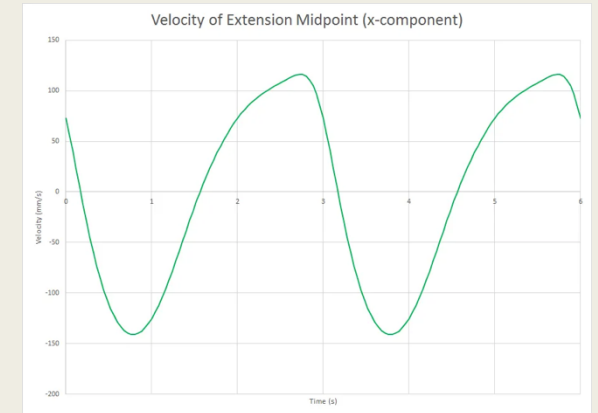
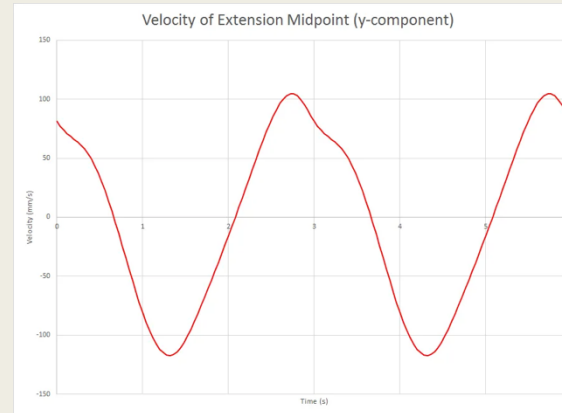
Motion studies enable collection of data across multiple performance metrics. While traditional hand calculations can be challenging due to complex physics equations, computer-aided analysis performs these calculations almost instantly.

In our study, we tracked the movement of a single point—specifically, the midpoint of the extension—throughout the simulation.

The resulting motion data was converted into graphical form for better visualization.

RESULT:

According to the motion study analysis, the maximum vertical forces on bearings 1 and 2 are 0.35 N and 0.17 N respectively. This information is valuable for determining if the assembly can withstand the forces generated by its motion.



Rocket Landing Trajectory Optimization

During Starship SN15's scrubbed launch window, I developed a 2D simulation of its landing maneuver.

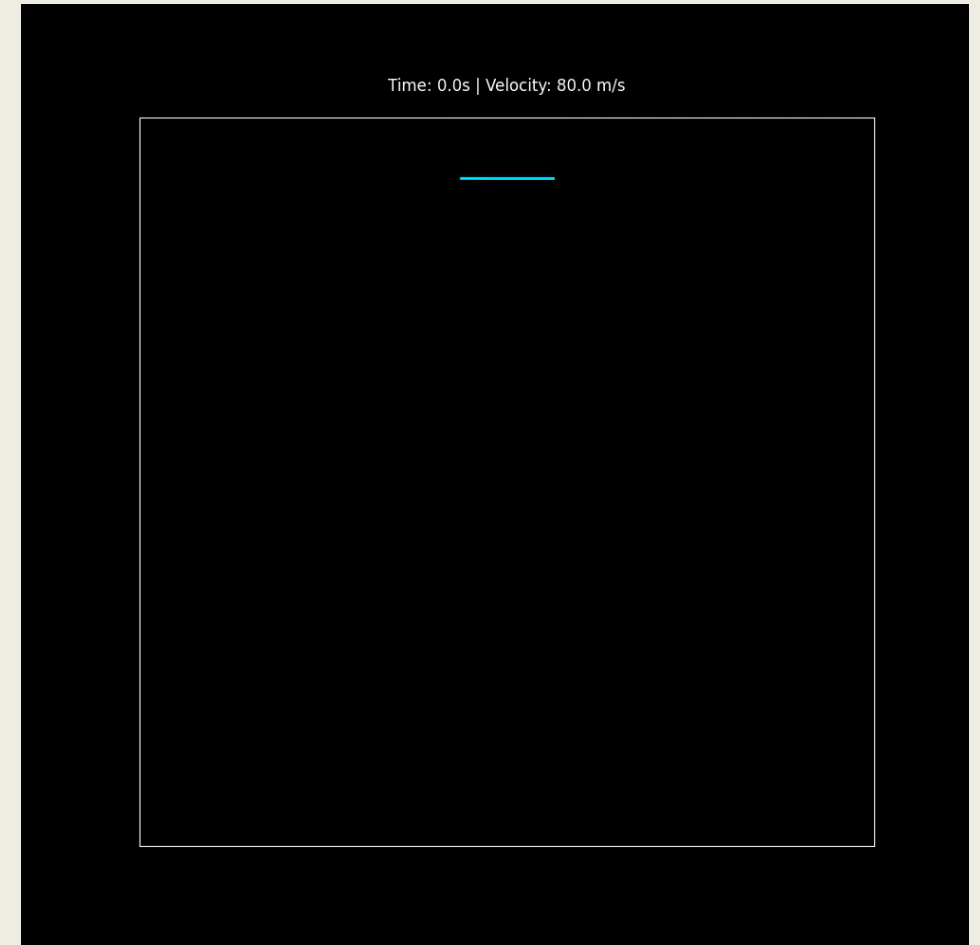
After optimizing the flight dynamics model, I recreated the vehicle's signature "belly flop" to vertical landing sequence.

Compared to actual SpaceX footage, the simulation showed remarkable accuracy - despite being developed purely from first principles without reference to empirical data.

To determine the best path, we define a "cost function" (like machine learning). A simple cost function might measure path length.

However, this can lead to impractical solutions - a shortest-path algorithm would ignore obstacles like a pit on the floor.

Click to play the video.



Pathfinder Robot

My team's approach to a proposed challenge to model the waiter mobile robot in MATLAB.

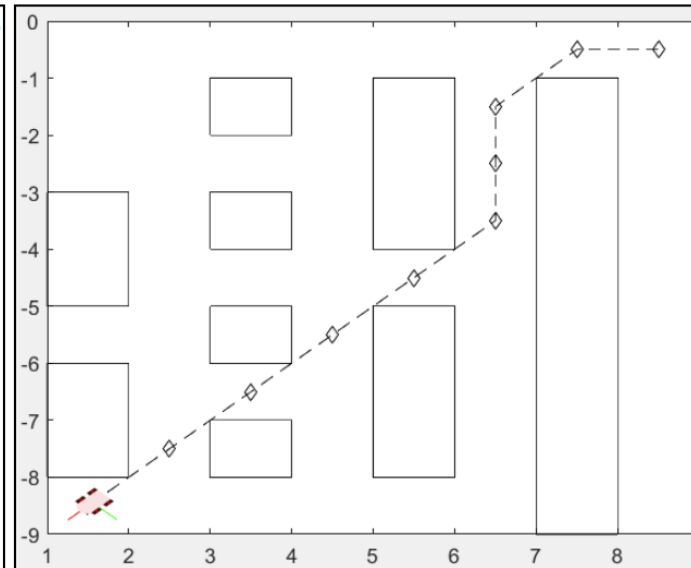
We developed an algorithm that enables a robot to navigate through a complex environment, avoiding obstacles and finding optimal paths to deliver items efficiently.

This project combined principles of robotics, path planning algorithms, and simulation techniques to create a realistic model of robot navigation in service environments.

Here's the [link](#) to the full presentation, report, and the MATLAB code.

```
%Define restaurant map. 0's are tables/walls
%The top left is defined as the origin
map = [ ...
    1, 1, 1, 1, 1, 1, 1, 1; ...
    1, 1, 0, 1, 0, 1, 0, 1; ...
    1, 1, 1, 1, 0, 1, 0, 1; ...
    0, 1, 0, 1, 0, 1, 0, 1; ...
    0, 1, 1, 1, 1, 1, 0, 1; ...
    1, 1, 0, 1, 0, 1, 0, 1; ...
    0, 1, 1, 1, 0, 1, 0, 1; ...
    0, 1, 0, 1, 0, 1, 0, 1; ...
    1, 1, 1, 1, 1, 1, 0, 1; ...
    ];

%Define a start and end (desired) node
desiredNode = [9,1];
startNode = [1,8];
```

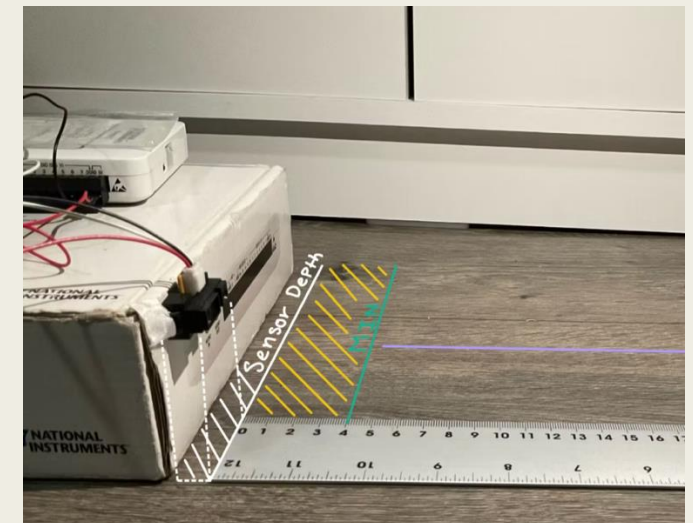
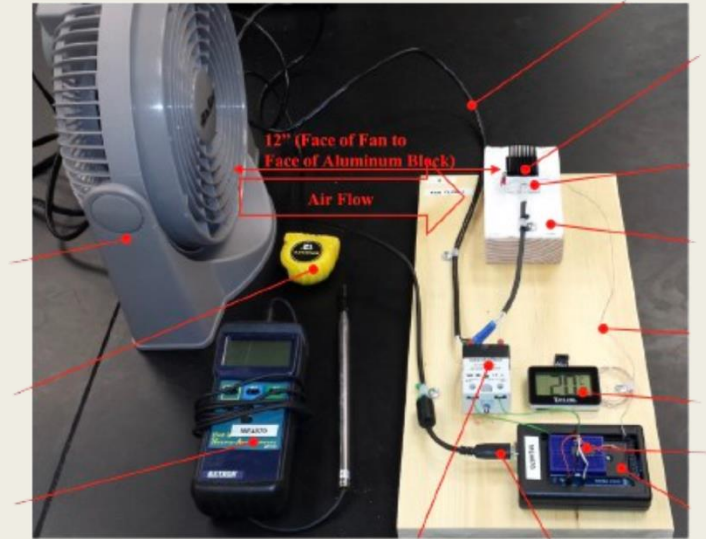


Sensor Calibrations

Over the course of my 2 measurement classes at Ohio State, I have experience working with and calibrating the following sensors. I have used NI9234 DAQ and LabView throughout these courses.

1. Thermocouples and Seebeck effect.
2. Pressure Transducer and Wheatstone Bridge.
3. PID Controllers.
4. Optical Sensor Calibration and designing an Automatic Hand-wash.
5. Accelerometers, Tachometers, and pressure gage to link kinematic attributes of the air compressor to frequency content in acceleration measurements on the air compressor.
6. Helmholtz resonators, comparing to theoretical and analyzing data using LabView
7. Thermal System Modelling with Thermocouples and Heat Sink.

Here's the [link](#) to look at all the final reports for the above-mentioned sensor calibrations and data analysis.



Research: AI-Driven Topology Optimization

Introduction

Topology Optimization (used to distribute material within a defined design space to maximize performance objectives) has emerged as a powerful computational tool in structural design since its introduction in the late 1980s [1]. Despite its proven success, widespread industry adoption faces two key challenges.

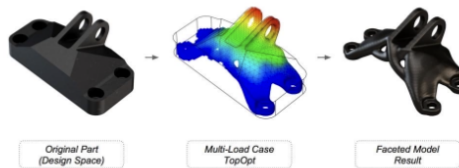


Figure 1: Topology Optimization Process

First, topology optimization remains **computationally intensive**, particularly for large-scale problems. Second, while the method generates theoretically optimal designs, these often result in **complex organic shapes** that pose manufacturing challenges and don't align with industry preferences for **simpler geometries** [2].

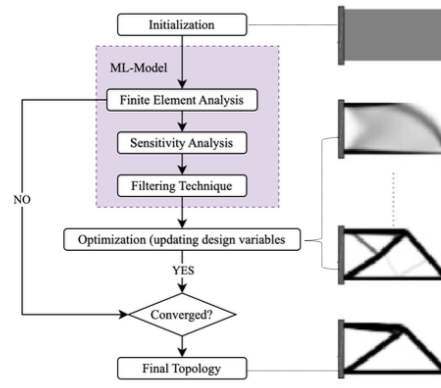


Figure 2: ML in Topology Optimization Loop

Objective

To develop an **AI-enhanced framework** that uses machine learning for **geometric primitive-based** topology optimization to accelerate mechanical design while maintaining high structural performance.

Methodology

Data Generation Collection

Our topology optimization dataset was created using the TopOpt library to generate diverse designs with varied parameters. The dataset includes MBB beams and cantilever beams with different mesh resolutions (120×60 to 200×100), volume fractions (0.2-0.7), and penalty factors.

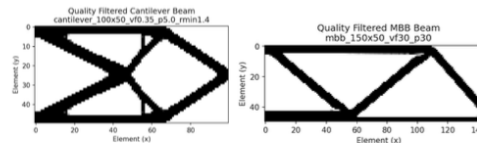


Figure 3: Sample Cantilever and MBB Beam Solved

Binary Processing and Feature Extraction

The binary processing pipeline transforms optimized density fields into meaningful structural representations through key stages: thresholding to convert continuous fields to binary images, noise removal to eliminate artifacts, edge detection to identify boundaries, and skeletonization to create centerline representations.

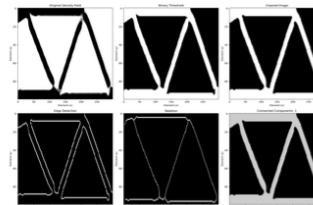


Figure 4: Feature Extraction Steps for MBB (in order)
Binary Processing, Cleanup, Edge Detection, Skeletonization

Results

Primitive Identification + Library

In our primitive identification process, we decompose structures into fundamental building blocks. The system detects junction nodes (classified as L, T, X, Y junctions) and identifies branches connecting these junctions (categorized as beams, struts, or connectors). This creates a structural library and enables graph-based representation of each optimized design as a network of interconnected elements.

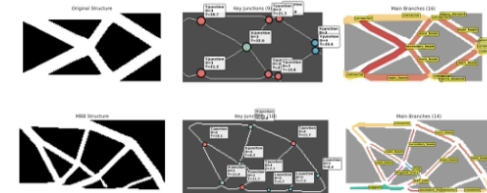


Figure 5: Junction and Branch Detection with Thickness

Machine Learning Model Prediction

Our model is purpose-built to detect only structural junctions (branch/merge nodes) in topology-optimized layouts. By restricting the task to junction recognition, we decouple connectivity from optimization and enable a clean, post hoc assembly step with fixed-thickness members.

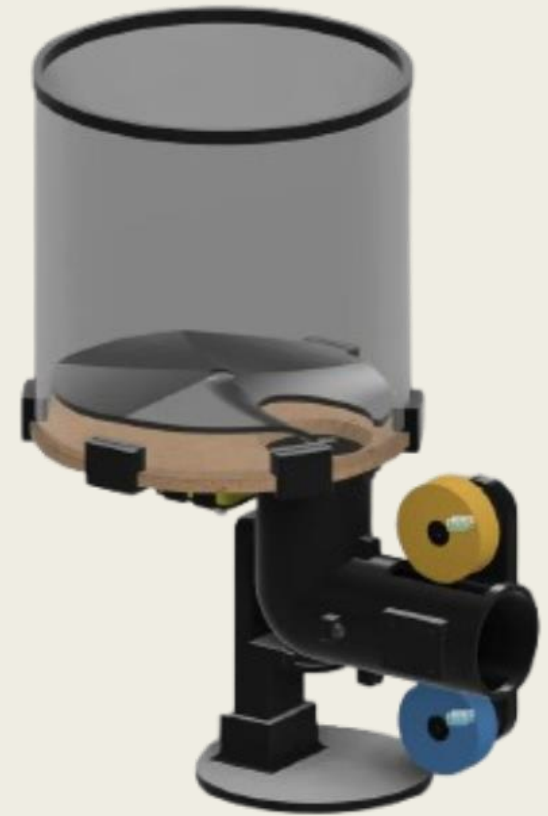
We trained on large, quality-filtered cantilever and MBB corpora spanning resolutions and parameters (volume fraction, penalization, filter radius). The model ingests primitive-driven features (binary, cleaned, edges, skeleton, components) to learn junction semantics that are stable across noise and scale.

In held-out designs, the model achieves high junction precision with tight spatial localization—very few false positives along straight members and consistent hits at true branch points. Recall is strong for typical junctions and moderately lower for ultra-thin or low-contrast cases. Performance remains robust across mesh refinements and varying filter radii, indicating good generalization.

Current Projects: Work in Progress

1. 6 Degree of Freedom Robotic Arm.
2. Table Tennis Ball Dispenser Robot.

Attached images are the first versions of the CAD.



Miscellaneous Software Projects

Some software projects I did for fun.

1. My personal portfolio website: eshwarp.com
2. Flight Delay Analysis: helping find the airline that gets delayed the least: howlateisyourflight.com
3. Website for my university's poker club: